UNIVERSITY OF THE PUNJAB

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Sixth Semester 2015
Examination: B.S. 4 Years Programme

Roll No. 0077?3

PAPER: Physical Chemistry
Course Code: CHEM-313

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions on separate answer sheets.

Questions with short answers.

Q.1- Give short comings of Boltzman's distribution Law. G ST (2) 269 (S)

Q.2- Derive sterling's approximation formula. (2) (263) ST

Q.3- What is the effect of molar mass on the vertical distribution of molecules according to Barometric distribution law? (2) (31) K

Q.4- Justify, Reaction in solution phase are faster than gas phase. (2) Notes

Q.5- Sketch the plot 'K' versus 'T' according to Arrhenius equation. Ck (2)

Q.6- Entropy of universe increases. Comment. (2) CT

Q.7- What is adiabatic demagnetization? Give Examples? (2) CT

Q.8- What is thermodynamic probability? (2) (262)

Q.9- Derive expression for average velocity using Max well's equation. (2) 23 (K)

Q.10- What is Nernst heat theorm. Explain briefly? (2) CT

Questions With Brief Answers 3×10=30

Q.8(a) What are unimolecular gas phase reactions? Ck (2)

(b) Give kinetics of unimolecular gas phase reactions Ck (8)

Q.9-(a) What is third Law of thermodynamics? How it is used for the determination of entropies. (10) CT

Q.10- Develop a relation between absolute entropy and partition function. (10) ST

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Sixth Semester 2016

Examination: B.S. 4 Years Programme

Roll No. ..10029.....

Physical Chemistry
Code: CHEM-313

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Section II (Subjective)

Q. 2 Answers the following short questions: (2×10=20)

(a) Define the term cage effect. Ck

(b) What are fast reactions? Ck

(c) Ideal gases do not exist but it is a useful concept. Comment. (2)(K)

(d) Give a mathematical relation between Gibbs energy, entropy and enthalpy of a process. (280) ST

(e) What is the effect of temperature on vertical distribution of gas? (31)(K)

(f) Describe term Clausius in-equality. CT

(g) Write two different forms of Arrhenius equation. Ck

(h) What is difference between Classical and Statistical thermodynamics? CT

(i) Give two applications of Barometric formula. (29)(K)

(j) What do you understand by unimolecular gas phase reaction? Ck

Q.3 (a) State and explain third law of Thermodynamics. CT (5)

(b) Give experimental verification of third law of Thermodynamics. CT (5)

Q.4 (a) What are various types of methods of kinetic study of fast reactions? Discuss relaxation method in detail. Ck (5)

(b) Give five postulates of transition state theory. Ck (5)

Q.5 (a) What is partition function? Give its significance ST (5)

(b) Discuss effect of molar mass and altitude on vertical distribution of particles. (5) 31

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2017
Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Physical Chemistry
Course Code: CHEM-313

TIME ALLOWED: 2 hrs. & 30 mins.
MAX. MARKS: 50

Attempt this Paper on Separate Answer Sheet provided.

Note: Attempt all questions.

Section II (Subjective)

Q. 2 Answers the following short questions: (2×10=20)

(a) Define the term partition function. ST

(b) What are ionic reactions? CK

(c) Entropy is state function. Comment. CT

(d) Give a mathematical relation between standard Gibbs energy and [illegible] constant. CT

(e) What is the effect of temperature on rate constant? CK

(f) What are ideal gases? Notes

(g) How can you determine pre-exponential factor from Arrhenius equation? CK

(h) What is difference between translational and rotational partition functions? ST

(i) Give significance of Barometric formula. 32

(j) What do you understand by cage effect? CK

Q.3 (a) State and explain Nerst heat Theorm. CT

(b) Derive two Maxwell relations. CT (5)

Q.4 (a) What is Eyring equation? How can you determine Eyring parameters [illegible] using Eyring equation? CK (5)

(b) Give five postulates of Collision theory. CK

Q.5 (a) What is partition function? Give its significance. ST

(b) Discuss effect of temperature and altitude on vertical distribution of particles. (5) (5) 31

UNIVERSITY OF THE PUNJAB

Sixth Semester - 2018

Examination: B.S. 4 Years Programme

Roll No. ....

PAPER: Physical Chemistry — **TIME ALLOWED: 2 Hrs. & 45 Mints.**

Course Code: CHEM-313 Part – II — **MAX. MARKS: 50**

Attempt this Paper on Separate Answer Sheet provided.

Q. 2 Answers the following short questions: (2×10=20)

(a) Define the term dynamic equilibrium.

(b) What is cage effect? Ch

(c) Reaction in solution phase are faster than reaction in gas phase. Comment. Notes

(d) Give a mathematical relation between standard Gibbs energy and equilibrium constant. ST

2022 (e) What is the effect of temperature on vertical distribution of gas? 31

(f) Describe partition function. 270 (S)

(g) ~~Write~~ two Maxwell relations. CT

(h) What is difference between spontaneous and non-spontaneous process? CT

(i) Give two applications of Barometric formula. 29

(j) What do you understand by ionic strength? Ch

Derive Maxwell's law of distribution of molecular velocities. 15-20 (10)

Describe Transition State theory. Derive Eyring equation to explain the effect of temperature on rate of reaction in solution. Ch (10)

(a) Discuss applications of partition function for evaluation of thermodynamic parameters. ST (5)

(b) Derive two Maxwell relations and give their significance. CT (5)

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2019

Roll No. ....

Paper: Physical Chemistry
Course Code: CHEM-313 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

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Q. 2 Answers the following short questions: (2×10=20)

(a) Define the term microstate. ST

(b) What are effective collisions? 6

2022 (c) Give two statements of 2nd law of thermodynamics. CT

(d) Give a mathematical relation between Gibbs energy, entropy and enthalpy of a process. ST

(e) What is the effect of temperature on vertical distribution of gas? 31

(f) Describe term Clausius in-equality. CT

2022 (g) Write two different forms of Arrhenius equation. CK

(h) What is difference between Classical and Statistical thermodynamics? CT

(i) Give mathematical formulation of Barometric formula. 29

(j) What do you understand by fast reactions? CK

Q.3 (a) State and explain 2nd law of Thermodynamics. CT (5)

(b) Discuss Clausius in-equality in Thermodynamics. CT (5)

Q.4 (a) Give five postulates of transition state theory. CK (5)

(b) Derive Eyring equation on the basis of postulates of transition state theory CK (5)

Q.5 (a) What is Sterling's approximation? Give its significance. ST (5)

(b) Discuss effect of molar mass and altitude on vertical distribution of particles. (5) 31

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – 2020

Roll No. ....

Paper: Physical Chemistry
Course Code: CHEM-313 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q. 2 Answer Short Questions. (10x2=20)

a) What is closed process? CT

b) What is partition function? ST

c) Define second law of thermodynamics. CT

d) What is meant by degree of freedom? ST

e) Describe most probable velocity 21

f) What is cage effect? Ch

g) Reactions in solution phase are faster than in gas phase. Why? Notes

h) What is law of conservation energy? CT

i) What is difference between spontaneous and non-spontaneous processes? CT

j) Explain Nernst heat theorem. CT

2022 k) What are closed and isolated system?

2022 l) Explain third law of thermodynamics?

2023 m) Write down expression for nernst approximation formula

Section III (Questions with Brief Answers) (3 x 10 = 30)

1. Define third law of thermodynamics and give its experimental verification. CT (10)
2. 2022 Derive and explain Maxwell's Distribution Law for distribution of molecular velocities among gas molecules. 15-20 (10)
3. (a) Discuss the effect of temperature on the rate of reaction on the bases of Arrhenius equation. Ch (5)

   (b) Derive Eyring equation using postulates of transition state theory. Ch (5)

UNIVERSITY OF THE PUNJAB

Roll No. ....

B.S. 4 Years Program / Sixth Semester – 2021

er: Physical Chemistry
urse Code: CHEM-313 Part – II

Time: 2 Hrs. 45 Min. Marks: 50

ATTEMPT THIS (SUBJECTIVE) ON THE SEPARATE ANSWER SHEET PROVIDED

Q.2. Answer the following short questions. (10x2=20)

a) Write mathematical relation between entropy and probability. ST

b) Give at least two postulates of bimolecular collision theory? ss CK

c) Derive average velocity? 23

d) What is side reaction? Give its type. CK

e) Describe most probable velocity. 21

f) What is cage effect? CK

g) Reactions in solution phase are faster than in gas phase. Why? Notes

h) Write expression of energy in term of partition function. ST

i) What is difference between spontaneous and non-spontaneous processes? CT

j) Write expression for Sterling Approximation ST

Q.3. Questions with brief answers. (3x10=30)

1. Derive various thermodynamical parameters in terms of translational partition function. ST (10)

2. (a) Explain bimolecular theory postulates? CK (5)
   (b) Verify third law of thermodynamics experimentally. CT (5)

3. (a) Effect of temperature and altitude on vertical distribution of particles? 31 (5)
   (b) Drive Eyring equation on the basis of transition state theory? CK (5)

UNIVERSITY OF THE PUNJAB

Roll No. ....

B.S. 4 Years Program / Sixth Semester – Spring 2022

Paper: Physical Chemistry — Course Code: CHEM-313 — Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(15x2=30)

Q.1. Write short notes on following.

2022 (a) What are ionic reactions? CK

(b) Describe barometric formula? 29

2022 (c) What is cage effect? CK

(d) How can you determine pre-exponential factor from Arrhenius equation? CK

2022 (e) What is the effect of temperature on rate constant? CK

2022 (f) What are effective collisions? 6

2022 (g) Give postulates of Bimolecular theory? CK

(h) Define partition function? What is factorization of partition function? ST

(i) Describe Nernst Heat Theorm. CT

2022 (j) Write down Nernst Approximation formula. CT

2022 (k) What is the difference between classical and statistical thermodynamics? CT

2022 (l) What is Clausius inequality? CT

2022 (m) What do you understand by fast reactions? CK

(n) Derive root means square velocity from Maxwell's distribution law? 24

2022 (o) What is pulse method? CK

Answer the following questions.

2022

Q.2 (a) What is Eyring equation? How can you determine Eyring parameters using Eyring equation? CK (5)

(b) What is the effect of temperature and altitude on vertical distribution constant? (5) 31

Q.3 (a) Derive expression of energy interms of "Partition Function". ST (5)

(b) Verify third law of thermodynamics experimentally. CT (5)

Q.4(a) Elaborate five postulates of transition state theory? CK (5)

(b) Derive two Maxwell's relations? CF (5)

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Roll No. ....

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2023

Time: 3 Hrs. Marks: 60

: Physical Chemistry Course Code: CHEM-313

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

(15x2=30)

.1. Answer the following short questions.

I How can you determine pre-exponential factor from Arrhenius equation.

II What is effect of temperature on rate constant?

II What are ionic reactions?

V What do you understand by cage effect?

V What are ideal gases?

VI Describe most probable velocity

VII Reactions in solution phase are faster than gas phase. Why?

VIII Derive root mean square velocity from Maxwell distribution law.

X Write down Nernst approximation formula.

Give two statements of 2^{nd} law of thermodynamics.

What is the effect of temperature on vertical distribution of gas?

What is meant by degree of freedom?

XI Give mathematical formulation of Barometric formula

XIV Describe term Clausis in-equality.

XV Define the term microstate.

Answer the following questions.

.2 (a) Derive expression for Sterling approximation.

(b) Prove that $Q = Q_t.Q_r.Q_v.Q_e$

.3 (a) State and explain 3^{rd} law of thermodynamics.

(b) Prove third law of thermodynamics experimentally.

.4 (a) Give five postulates of transition state theory.

(b) Derive Erying equation on the basis of postulates of transition state theory.

UNIVERSITY OF

B.S. 4 Years Program / Sixth Semester – Fall 2023

Course Code: CHEM-313

Time: 3 Hrs. Marks: 60

Paper: Physical Chemistry

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

I Explain spontaneous and non-spontaneous processes.

II Describe the most probable velocity.

III How can you determine the pre-exponential factor from the Arrhenius equation/plot?

IV What is the difference between isolated and closed systems?

V Define open system.

VI Give two statements of the 2^{nd} law of thermodynamics.

VII What is the "Partition Function"?

VIII How do ionic reactions differ from covalent reactions?

IX Define the term microstate.

X Write two different forms of the Arrhenius equation.

XI What is adiabatic demagnetization?

XII How does the barometric formula describe the variation of atmospheric pressure with altitude?

XIII What are fast reactions?

XIV What factors influence the rate of a bimolecular reaction?

XV How does the concept of entropy relate to the second law of thermodynamics?

Answer the following questions. (3x10=30)

Q.2a Write a note on Maxwell's law of distribution of velocities.

b Prove that $Q = Q_t.Q_r.Q_v.Q_e$

Q.3a State and explain different statements of the 3^{rd} law of thermodynamics.

b *Drive statistically expression for equilibrium constant using Partition Function.*

Q.4a *Write a note on transition state theory.*

b *What is the effect of temperature on the rate of reaction?*

UNIVERSITY OF THE PUNJAB

B.S. 4 Years Program / Sixth Semester – Spring 2024

Roll No. ....

Paper: Physical Chemistry

Course Code: CHEM-313

Time: 3 Hrs. Marks: 60

THE ANSWERS MUST BE ATTEMPTED ON THE ANSWER SHEET PROVIDED

Q.1. Answer the following short questions. (15x2=30)

I Explain the third law of thermodynamics.

II What are ionic reactions?

III Explain the difference between a closed system and an isolated system.

IV What do you understand by the cage effect?

V Define average velocity. Determine its expression.

VI Describe most probable velocity.

VII Reactions in solution phase are faster than gas phase. Why?

VIII Derive root mean square velocity from Maxwell distribution law.

IX Define state function.

X Give two statements of 2nd law of thermodynamics.

XI Draw and label Carnot's cycle.

XII What is an open system?

XIII What factors influence the rate of a bimolecular reaction?

XIV What is the effect of temperature on the vertical distribution of gas?

XV How can one determine the pre-exponential factor from the Arrhenius equation.

Q.2. Answer the following questions. (3x10=30)

1. Derive four Maxwell's relations.
2. What is the Erying equation? How can one determine Erying parameters using the Eyring equation?
3. a) Give five postulates of transition state theory.
   b) Derive an expression for Maxwell's law of distribution of velocities.